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Business, Economic and Financial Data Final Project

A Time Series Analysis of Cinema Admissions in Italy

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Academic Year:
2024/2025

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1 Introduction

Over the past few years, we have grown accustomed to watching movies on OTT (Over-The-Top) streaming—services that deliver content directly to viewers over the internet. These platforms have become an integral part of our daily routines, allowing us to enjoy movies in the comfort of our homes. However, this shift has made us forget the unique joy of sharing emotions with a crowd in a movie theater.

Particularly after the COVID-19 pandemic, it seems like the streaming platforms became dominant, and the overwhelming volume of content on these services has made it increasingly difficult for individual films to draw attention in cinemas. This led me to wonder: is traditional cinema culture approaching its end?

To explore further, this study focuses on the time series analysis and forecasting of cinema admissions in Italy.

1.1 Problem Definition

This study aims to address the following questions:

- Is traditional cinema culture coming to an end?
- Will OTT services dominate the distribution of movies in the future?
- Have the COVID-19 pandemic and/or OTT platforms impacted theater admissions significantly?

1.2 The Data Sources

The dataset used in this study consists of monthly data spanning from January 2006 to December 2023, complemented by a simplified annual dataset that includes only OTT Subscriptions and OTT Spending.

The data sources are as follows:

- **Società Italiana degli Autori ed Editori (SIAE):**
  Number of screenings in theaters, cinema admissions, ticket prices, audience spending [1].
- **International Video Federation (IVF) & Statista:**
  OTT subscriptions and OTT spending (annual data only) [4, 5].
- **Istituto Nazionale di Statistica (ISTAT):**
  Consumer Price Index (CPI) (Month-over-Month) for all items and CPI (MoM) for recreation and culture [3].
- **Istituto Superiore di Sanità (ISS):**
  Number of reported COVID-19 cases [2].
- **Google Trends:**
  Search interest for OTT platforms, used as a proxy for streaming consumption behavior [7].

2 Data Exploration & Preprocessing

2.1 Dataset Overview

This study uses two datasets: a monthly dataset (January 2006 - December 2023) and an annual dataset (2006 - 2023). These datasets provide detailed insights into cinema admissions, audience spending, economic indicators, and trends related to online streaming services (Netflix, Amazon Prime Video, etc.).

2.1.1 Monthly Dataset

The monthly dataset serves as the primary source for time series modeling. The key variables included are:

- **Cinema Admissions:** The number of individuals attending cinemas each month.
- **Number of Screenings:** The total number of movie screenings conducted per month.

- **Cinema Audience Spending:** The total expenditure by cinema-goers, including ticket sales and other expenses (in euros).
- **Average Spending Per Person:** The average amount spent per visitor in a given month (in euros).
- **Google Trends OTT:** A measure of online search interest in OTT (Over-the-Top) platforms. According to Statista's study on video-on-demand platforms, the most popular SVOD services in Italy as of June 2023 was Netflix, second came Amazon Prime Video, and Disney+ followed behind. [6] Therefore Google Trends for these platforms serve as an estimate of consumer engagement with streaming services.
- **CPI (All Items):** The month-over-month Consumer Price Index for all goods and services, reflecting general inflation.
- **CPI (Culture):** The CPI specific to cultural and entertainment-related expenses.
- **COVID-19 Cases:** The number of reported COVID-19 cases per month.

2.1.2 Annual Dataset

The annual dataset provides OTT Subscriptions, OTT Spending in annual format and aggregated yearly data on cinema attendance. This dataset is used to examine long-term competition in cinema admissions and OTT subscriptions. The key variables included in the annual dataset are:

- **OTT Subscriptions:** The total number of active OTT platform subscriptions per year.
- **OTT Spending:** The total consumer expenditure on OTT services (in euros).
- **Cinema Admissions:** The yearly total of cinema admissions.

2.2 Handling Missing Values

The dataset initially lacked monthly data for cinema-related features (audience admissions, number of screenings, audience spending, average spending per person) for the year 2013, with only the total annual values available. To address this issue, monthly values were imputed using a proportion-based approach to ensure consistency with historical trends and seasonality. To impute the missing values, monthly averages from the adjacent years (2011, 2012, 2014, and 2015) were used to calculate proportional distributions. These proportions were then applied to the total annual values for 2013.

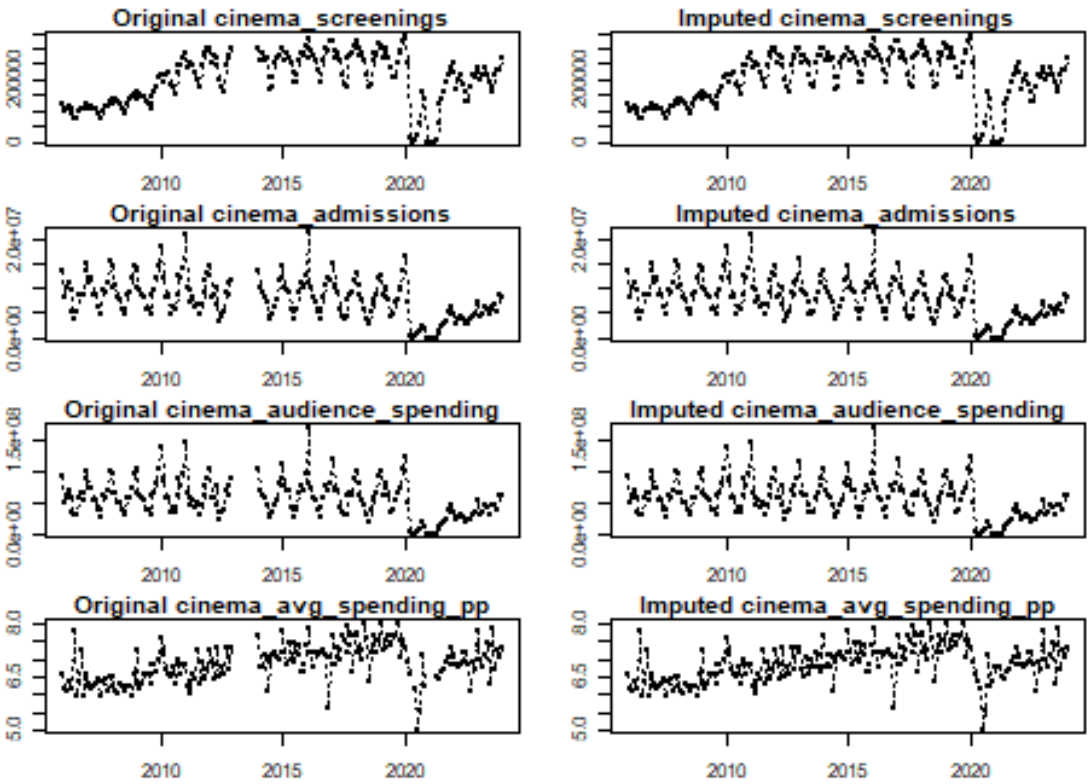

Figure 1: Original vs. Imputed Monthly Dataset

Additionally, a small number of missing values in 2020 and 2021 (likely due to disrupted operations during the COVID-19 period lock-downs) were replaced with the average ticket price for those respective years.

The annual dataset doesn't have missing values.

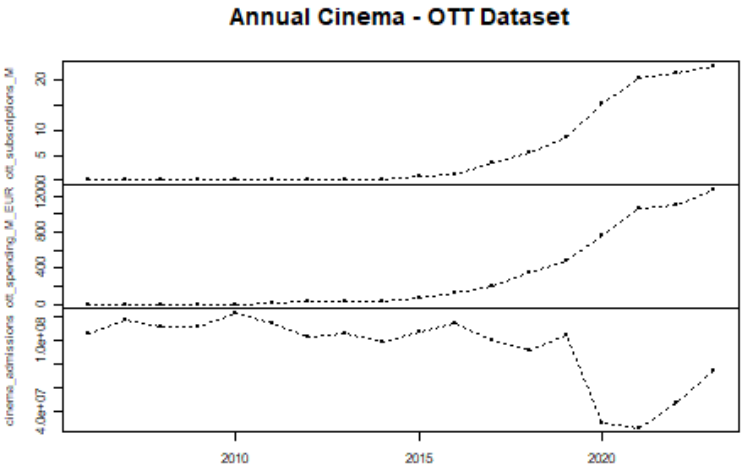

Figure 2: Overview of the Annual OTT Dataset

2.3 Collinearity Check

Before proceeding with modeling, it is important to examine potential collinearity issues among the predictor variables.

2.3.1 Correlation Analysis for Monthly Data

To assess the degree of linear relationships between variables, a correlation matrix was computed for the monthly dataset.

- **Cinema Admissions and Audience Spending** exhibit a very strong positive correlation.
- **Cinema Admissions and Number of Screenings** are also positively correlated.
- **Google Trends OTT and Cinema Admissions** display a moderate negative correlation.
- Inflation Indicators (CPI All Items, CPI Culture) show weak correlations with other variables.

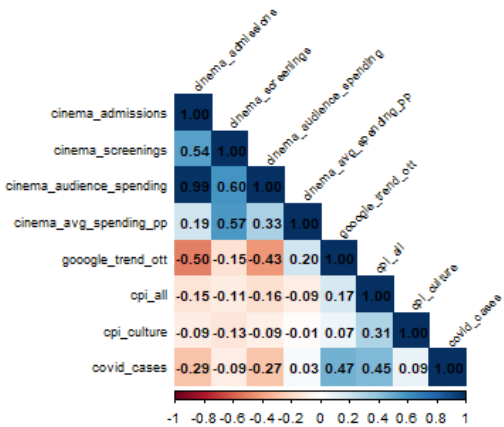

Figure 3: Correlation Matrix for Monthly Data

2.3.2 Variance Inflation Factor (VIF) Analysis for Monthly Data

While the correlation matrix provides valuable insights, it does not explicitly indicate whether collinearity poses a problem for regression models. Therefore, VIF analysis was conducted.

All VIF values are below 5, suggesting that multicollinearity does not pose a major issue for regression modeling. Cinema Screenings has the highest VIF value (2.13), followed by Cinema Audience Spending (1.98), but both remain well within the acceptable range.

Variable	VIF Value
Cinema Screenings	2.13
Cinema Audience Spending	1.98
Cinema Avg Spending Per Person	1.77
OTT Google Trends	1.75
CPI (All Items)	1.39
CPI (Culture)	1.13
COVID-19 Cases	1.58

Table 1: Variance Inflation Factor (VIF) for Monthly Dataset

2.3.3 OTT Google Trends as a Proxy for OTT Subscriptions and Spending

Since OTT Subscriptions and OTT Spending are only available in annual format, this section examines whether Google Trends OTT data can serve as a proxy for monthly modeling.

The correlation matrix presented in Table 2 confirms a strong relationship between the variables.

The results strongly suggest that Google Trends OTT search activity closely follows OTT Subscriptions and Spending trends over time. This supports the assumption that Google Trends data can serve as a reliable proxy for modeling OTT behavior. The visualization suggest that search interest in OTT platforms generally moves in sync with actual market

Variable Pair	Correlation Coeff.
OTT Subscriptions vs. OTT Spending	0.9981
OTT Subscriptions vs. Google Trends OTT	0.9379
OTT Spending vs. Google Trends OTT	0.9344

Table 2: Correlation Between Google Trends OTT and OTT Data

adoption, though there may be a lagged relationship where increases in search activity trigger corresponding rises in subscriptions and spending.

Based on the high correlation values and trend alignment, Google Trends OTT data can be considered a viable proxy for monthly OTT Subscriptions and Spending.

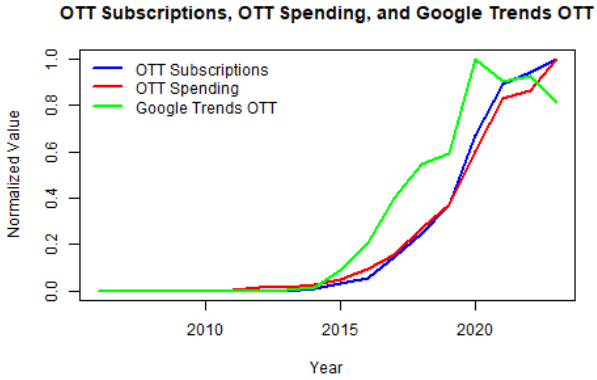

Figure 4: Comparison of OTT Subscriptions, OTT Spending, and Google Trends OTT (Normalized)

2.4 Time Series Decomposition

Before testing models on the data, an in-depth analysis of the Cinema Admissions time series is performed to examine trend, seasonality, and any possible disruptions.

2.4.1 Time Series and Autocorrelation Analysis

To have an understanding of the data, the raw time series and its correlograms are plotted in Figure 5.

- The time series exhibits strong seasonality
- A downward trend is visible after 2020, indicating a structural break caused by the COVID-19 pandemic.
- The correlogram shows significant autocorrelations at seasonal lags, confirming the presence of seasonality.

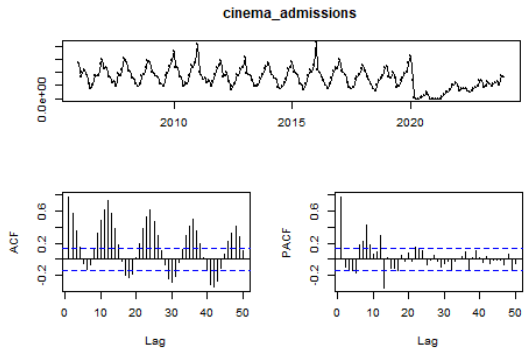

Figure 5: Time Series and Autocorrelation Analysis for Cinema Admissions

2.4.2 Seasonal Pattern Analysis

Cinema admissions data reveals distinct seasonal behavior, visualized in Figure 6.

- Admissions exhibit a recurring seasonal cycle, with peaks in January and December (likely due to holiday season).
- Lowest admissions are observed during summer season.
- The years 2020 and 2021 deviate significantly from seasonal behavior, indicating the COVID-19 impact.
- The recovery in 2022 and 2023 suggests a recovery, but levels remain lower than earlier years.

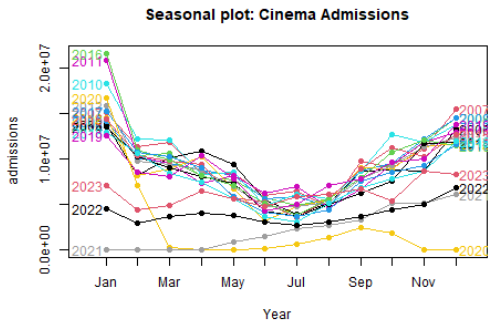

Figure 6: Seasonal Plot: Cinema Admissions (Yearly Comparison)

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